Fix 2: Removing the Constant from the `msj` Energy

autotwin/automesh issue #768

October 14, 2026

Summary

Fix 2 subtracts $\Theta\,\ell^2$ from every hex's term in the `conspire msj` energy. It does not change any step that L-BFGS takes. It changes only the denominator of the stall test. On the snapped bone octree at $\Theta = 0.32$, that denominator falls from 6,353.7 to 1.19×10^{-3}. The stall test then reads full progress as full progress, and the ratchet continues from 0.319 to 0.470.

Notation

h is one hex, and $\ell(h)$ is its average edge length. $\hat{J}(h)$ is the minimum scaled Jacobian of h over its eight corners and its body center. $\Theta \in [0, 1]$ is the current ratchet threshold for the minimum scaled Jacobian $\hat{J}$. Each hex adds one term $\psi(h) = \ell^2 \min(\hat{J}, \Theta)$, the `conspire` form ψ_c of the companion HexOpt note. The paper writes ψ as $ReHQJ$, $\hat{J}$ as SJ, ℓ as $\bar{e}$, and $\mathcal{L}$ as L. `improve` minimizes the augmented Lagrangian $\mathcal{L} = -\sum_h \psi(h)$ plus the terms that hold the boundary on the STL. $\langle x \rangle = \max(x, 0)$ is the Macaulay bracket. Within one ratchet level, Θ and every ℓ are fixed.

1. The stall test is a ratio

After each L-BFGS run, `improve` computes

$$\text{ratio} = \frac{\mathcal{L}_\text{before} - \mathcal{L}_\text{after}}{|\mathcal{L}_\text{before}|} \tag{1}$$

and stops the level if the ratio is below $\texttt{STALL} = 10^{-6}$. The numerator measures the work that the run did. The denominator is meant to measure the work that remains.

2. The constant inflates only the denominator

As shipped, a hex with $J > 0$ adds

$$\psi(h) = \ell^2 \min\big(\hat{J}(h), \Theta\big). \tag{2}$$

Above Θ, the term is the constant $\Theta\,\ell^2$. No node move changes it, so it contributes nothing to the numerator. It still sits in the denominator.

At $\Theta = 0.32$ on the octree, 9,266 hexes sit above Θ. Their constant terms sum to 6,353.4. One hex sits below Θ, and its term is 0.312. The worst element is 0.3188, so its scaled Jacobian needs to rise by 0.0012 to meet Θ.

Table 1 gives the first L-BFGS run at this level, from the debug log. The run lowers $\mathcal{L}$ by 1.19×10^{-3}, which is the full shortfall of the one hex. Divided by 6,353.7, that drop gives a ratio of 1.9×10^{-7}. The ratio is below 10^{-6}, so `improve` stops the level.

3. Fix 2 removes the constant

Fix 2 subtracts $\Theta\,\ell^2$ from every term. Written as a per-hex energy that $\mathcal{L}$ sums directly, the result is

$$w(h) = \Theta\,\ell^2 - \psi(h) = \ell^2\,\big\langle \Theta - \hat{J}(h) \big\rangle, \qquad \mathcal{L} = \sum_h w(h) + \text{boundary terms.} \tag{3}$$

w is nonnegative. A hex above Θ adds exactly 0. A hex below Θ adds its shortfall below Θ, weighted by ℓ^2. The $J \le 0$ case shifts by the same constant, to $w = \Theta\,\ell^2 - J/\ell$. Figure 1 shows both forms of one hex's contribution to $\mathcal{L}$. The two curves are parallel, offset by $\Theta\,\ell^2$.

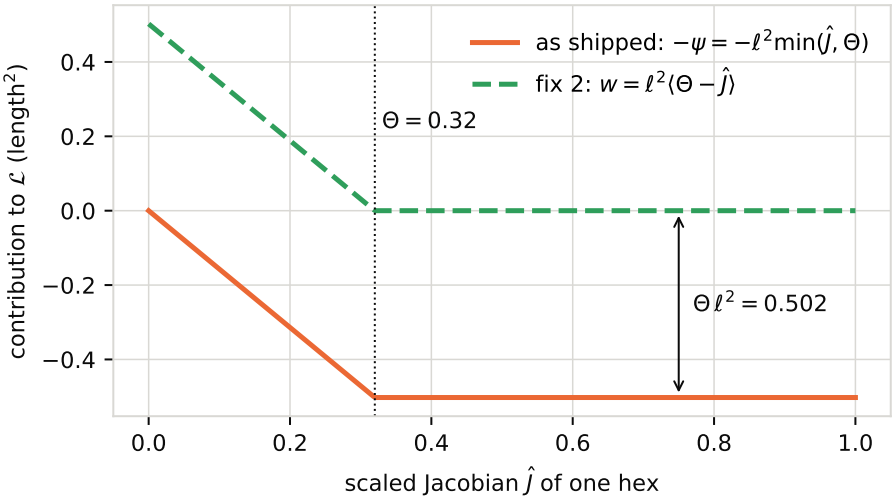


Figure 1: One hex's contribution to $\mathcal{L}$, with $\ell = 1.253$, the median on the snapped octree, at $\Theta = 0.32$. As shipped (solid), $-\psi$ is flat at $-\Theta\,\ell^2 = -0.502$ above Θ. With fix 2 (dashed), w is flat at 0. Both have the same slope everywhere.

The hex part of $\mathcal{L}$ is now the total shortfall of all hexes below Θ. At $\Theta = 0.32$, the 6,353.4 disappears. The same run now starts at $\mathcal{L} = 1.19 \times 10^{-3}$ and ends at 1.7×10^{-24}. The ratio is 1.000, far above 10^{-6}, so the level continues.

first L-BFGS run at $\Theta = 0.32$	as shipped	fix 2
worst element at start	0.3188	0.3188
$\mathcal{L}_{\text{before}}$	−6,353.693	1.186×10^{-3}
drop $\mathcal{L}_{\text{before}} - \mathcal{L}_{\text{after}}$	1.19×10^{-3}	1.19×10^{-3}
ratio	1.9×10^{-7}	1.000
outcome	stops	continues

Table 1: The same run, before and after fix 2, from the `msj` debug logs on the snapped octree. The drop matches to three digits.

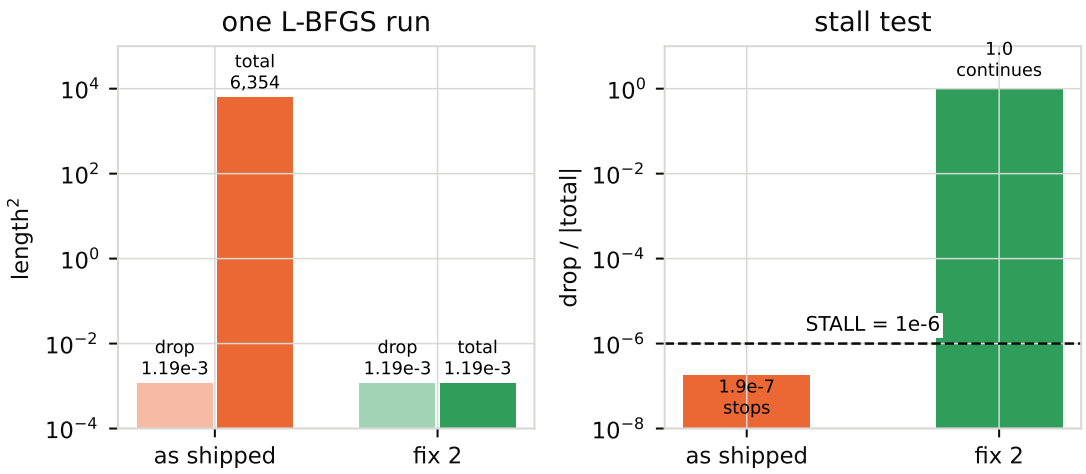


Figure 2: Left: the drop and $|\mathcal{L}_{\text{before}}|$ for the run in Table 1, on a log scale. The drop is the same, and only the total changes. Right: the ratio against `STALL` $= 10^{-6}$ (dashed).

4. The steps do not change

Within one level, $\Theta\,\ell^2$ is the same number before and after every node move. Subtracting it leaves two things unchanged.

- **The gradient.** The derivative of a constant is zero, so $\nabla\mathcal{L}$ is the same. L-BFGS picks the same search direction.
- **Every energy difference.** The constant cancels in $\mathcal{L}(\text{new}) - \mathcal{L}(\text{old})$. The Armijo condition compares exactly this difference, so it accepts the same step lengths.

L-BFGS therefore takes the same path. The drop in Table 1 confirms this: 1.19×10^{-3} in both columns. Only the stall test sees a different number.

5. The effect on the ratchet

Figure 3 shows the worst element at each accepted level on the octree. As shipped, the ratchet accepts 2 levels and stops at 0.319. With fixes 1 and 2, it accepts 18 levels and reaches 0.470. It then stops at $\Theta = 0.48$ after all 20 sweeps of that level. That limit is separate from the stall test.

6. Why this matters beyond one mesh

- **The constant grows with the mesh.** Each hex above Θ adds $\Theta\,\ell^2$. A mesh with ten times the hexes carries about ten times the constant. Lowering `STALL` to 10^{-9} moves the octree past 0.32, but a larger mesh would bring the stall back. Fix 2 removes the dependence on mesh size.
- **The energy has a direct meaning.** After fix 2, $\mathcal{L} = 0$ exactly when every hex meets Θ. The value of $\mathcal{L}$ shows how much work remains.

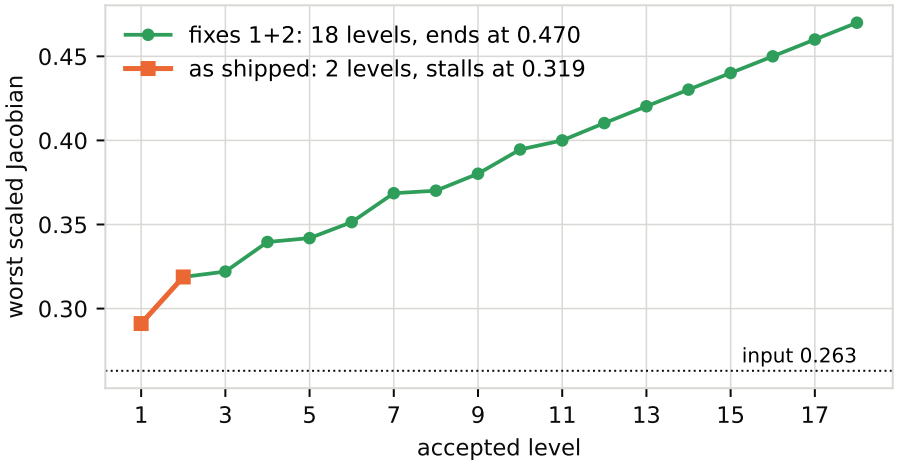


Figure 3: Worst scaled Jacobian at each accepted ratchet level on the snapped octree, from the debug logs. The dotted line is the input, 0.263.

7. What fix 2 does not address

As the shortfall approaches 0, the ratio divides by a small number. In the six runs on 2026-10-14, five of six ended by using all 20 sweeps of a level, not by the stall test. These runs did not show the small-denominator case.